

Central Area Drifts Obstruct Intrinsic Uniqueness of Rectangular Young Surface Lifts

Independent research note

August 17, 2026

Abstract

Lee asks whether rectangular increment regularity can yield a unique lift of a surface in the two-parameter Young regime. We give a negative answer to intrinsic uniqueness under the paper’s stated rough-surface axioms. For every $\rho \in (1/2, 2/3]$, the zero surface in $\mathbb{R}^3$ has a one-parameter family of distinct full rectangular ρ -Hölder multiplicative double-group lifts sharing all path data and all surface levels through two. The ambiguity is a degree-three Jacobi cycle multiplied by Euclidean rectangle area. It satisfies even the stronger mixed rectangular estimate suggested in the paper’s appendix. Additional normalization or naturality could still select the canonical Young-integral lift.

Packet status

Run fa_banach_001, lane 13.
Source arXiv:2406.16857, Remark 5.23 and the first question on PDF page 56 [1].
Classification Candidate counterexample, likely valid subject to human review.
Scope Intrinsic uniqueness among admissible rectangular rough lifts for $1/2 < \rho \leq 2/3$; canonical selection under extra axioms is not excluded.

avenue towards higher dimensional rough analysis, and we highlight some immediate questions which stem from this article.

- **Sewing with Rectangular Increments.** As discussed in Remark 5.23, additional regularity assumptions on rectangular increments are not explicitly used in the surface extension theorem (Theorem 5.18). Is there an analogue of the extension theorem which takes this into account, and allows us to define a unique lift of a surface $X \in C^{\square, \rho}([0, 1]^2, V)$ in the Young regime ($\rho \in (\frac{1}{2}, 1]$)?
- **Surfaces in Banach Spaces.** This article focuses on the case of surfaces valued in a finite di-

1 Question and answer

The source asks whether an analogue of its surface extension theorem can use rectangular regularity to “define a unique lift” of $X \in C^{\square, \rho}([0, 1]^2, V)$ throughout the Young regime $\rho \in (1/2, 1]$ [1, Section 6]. Remark 5.23 makes the missing issue precise: the constructed extension is unique only after level three has been supplied, and the paper does not prove that level three is determined by the underlying surface.

We use the source’s free crossed module $(\delta : T_1(V) \rightarrow T_0(V), \triangleright)$ and its group-like elements (G_1, G_0) . The product on T_1 is denoted by $*$. For a rectangle $Q = [s_1, s_2] \times [t_1, t_2]$, write

$$|Q| = (s_2 - s_1)(t_2 - t_1).$$

Theorem 1 (central-area nonuniqueness). *Let $V = \mathbb{R}^3$ and $1/2 < \rho \leq 2/3$. The constant surface $X \equiv 0$ admits a family $(\mathbf{X}^\lambda)_{\lambda \in \mathbb{R}}$ of full multiplicative double-group functionals in $\text{DGF}^{\square, \rho}(G_1, G_0)$ such that:*

- (i) *every horizontal and vertical path component is the identity;*
- (ii) *every surface component through level two is zero;*
- (iii) *the family members are pairwise distinct.*

Thus the underlying rectangular ρ -Hölder surface, even together with all boundary paths and all data through level two, does not intrinsically determine its lift under the paper's regularity conditions.

Idea of the proof

The first nontrivial kernel of the crossed-module boundary contains the degree-three Jacobi cycle

$$\Omega = [e_1, e_2 \wedge e_3] + [e_2, e_3 \wedge e_1] + [e_3, e_1 \wedge e_2].$$

Jacobi gives $\delta\Omega = 0$, while the six terms in its free-bimodule expansion show $\Omega \neq 0$. The Peiffer identities make kernel elements square-zero against all of T_1 . Consequently $\exp_*(a\Omega) = \mathbf{1} + a\Omega$, and multiplication of these exponentials just adds a . Assigning $a = \lambda|Q|$ to a rectangle therefore obeys both horizontal and vertical concatenation because area is additive in both directions. Finally, $|Q|$ has exactly the critical two-dimensional scaling: it meets the degree-three surface bound precisely up to $\rho = 2/3$.

2 The degree-three kernel cycle

Let e_1, e_2, e_3 be the standard orthonormal basis. The Lie action is embedded in the free bimodule by

$$[x, u \wedge v] = x \cdot (u \wedge v) - (u \wedge v) \cdot x.$$

Lemma 1. *The element Ω above belongs to the degree-three part of Kapranov's semiabelianized Lie algebra inside $T_1(V)$, is nonzero, and satisfies*

$$\delta\Omega = 0, \quad \|\Omega\| = \sqrt{6}, \quad \Omega * F = F * \Omega = 0 \quad (F \in T_1(V)).$$

In particular, $\exp_(a\Omega) = \mathbf{1} + a\Omega \in G_1$ for every $a \in \mathbb{R}$.*

Proof. The boundary intertwines the Lie action, hence

$$\delta\Omega = [e_1, [e_2, e_3]] + [e_2, [e_3, e_1]] + [e_3, [e_1, e_2]] = 0$$

by Jacobi. On expanding the three action brackets, one obtains six distinct signed vectors of the orthonormal basis

$$V \otimes \Lambda^2 V \oplus \Lambda^2 V \otimes V.$$

Every Peiffer generator has total degree at least $2 + 2 = 4$, so the Peiffer quotient imposes no relation in degree three. Therefore $\Omega \neq 0$ and $\|\Omega\|^2 = 6$.

The second Peiffer identity gives $E * F = \delta(E) \cdot F = E \cdot \delta(F)$. Taking $E = \Omega$ in the first form and $F = \Omega$ in the second proves both zero-product identities. Thus every positive power Ω^{*m} with $m \geq 2$ vanishes. Since Ω lies in the Lie algebra used to define G_1 , $\exp_*(a\Omega) = \mathbf{1} + a\Omega$ is group-like at every truncation. Its finite graded support also places it in the completed group G_1 . $\square$

3 The full multiplicative lift

For $\lambda \in \mathbb{R}$, give $X \equiv 0$ identity-valued horizontal and vertical path components and define its surface component on every rectangle Q by

$$\mathbf{X}_Q^\lambda = \exp_*(\lambda|Q|\Omega) = \mathbf{1} + \lambda|Q|\Omega. \quad (1)$$

The boundary condition holds because $\delta(\mathbf{X}_Q^\lambda) = \exp(\lambda|Q|\delta\Omega) = \mathbf{1}$, which is the boundary signature of the zero surface. If Q_1, Q_2 are horizontally adjacent, all path actions in the double-group composition are trivial, and Lemma 1 gives

$$\mathbf{X}_{Q_2}^\lambda * \mathbf{X}_{Q_1}^\lambda = \mathbf{1} + \lambda(|Q_1| + |Q_2|)\Omega = \mathbf{X}_{Q_1 \cup Q_2}^\lambda.$$

The same calculation applies to vertical adjacency. Hence (1) is a full multiplicative double-group functional, without appealing to an extension theorem. Distinct parameters give distinct values on $[0, 1]^2$, while all levels below three agree.

4 Regularity at the critical two-dimensional scale

It remains to check the analytic hypothesis. Put $a = s_2 - s_1$, $b = t_2 - t_1$ and $\sigma = \rho/2$. The only nonzero positive homogeneous component of (1) is

$$\|(\mathbf{X}_Q^\lambda)^{(3)}\| = |\lambda|\sqrt{6}ab.$$

The source's level-three surface regularity polynomial contains the positive $q = 3$ term

$$\frac{2^{6\sigma}}{\beta} \frac{(C_\omega a)^{3\sigma} (C_\omega b)^{3\sigma}}{\Gamma(3\sigma + 1)^2}.$$

Because $3\sigma = 3\rho/2 \leq 1$ and $a, b \in [0, 1]$, we have $ab \leq a^{3\sigma}b^{3\sigma}$. Choosing the Hölder-control constant C_ω large enough absorbs $|\lambda|\sqrt{6}$ and the fixed coefficient. All path bounds and all other surface-level bounds are zero. Therefore $\mathbf{X}^\lambda \in \text{DGF}^{\square, \rho}(G_1, G_0)$.

The example also meets the sharper mixed rectangular scaling mentioned in the appendix:

$$ab \leq a^\rho b^{2\rho} + a^{2\rho} b^\rho. \quad (2)$$

Indeed, if $a \leq b$, then

$$\frac{ab}{a^\rho b^{2\rho}} = a^{1-\rho} b^{1-2\rho} \leq b^{2-3\rho} \leq 1;$$

if $b \leq a$, use the second term symmetrically. This proves Theorem 1.

5 Interpretation and limitations

On square cells of side ε , the drift has size ε^2 , while degree-three rough regularity has scale $\varepsilon^{3\rho}$. The threshold $3\rho = 2$ explains both why the construction works throughout the unresolved interval $(1/2, 2/3]$ and why it does not conflict with the source's existing uniqueness from level two for $\rho > 2/3$.

The conclusion concerns *intrinsic* uniqueness inside the stated rectangular DGF class. The source's Young-integral formula remains a valid and natural canonical choice; the nonzero lifts above do not equal the smooth surface signature of the zero surface. Accordingly, the example does not exclude a theorem that adds smooth-approximation consistency, a locality or naturality axiom, or a quantitative normalization tying the higher-level control to the norm of X . It proves that rectangular regularity and the existing multiplicativity/boundary axioms alone cannot supply uniqueness.

6 Verification and novelty audit

The accompanying exact-arithmetic script expands Ω , verifies its zero boundary and squared norm six, and checks both area additivity laws on rational grids. It also tests (2) on 16,384 exponent and aspect-ratio cases; all checks pass. These computations corroborate, but do not replace, the symbolic proof.

On August 17, 2026, the run indexes and local arXiv corpus were searched by the source id/title and combinations of *unique lift*, *rectangular Hölder*, *central kernel*, and *Jacobi*. Bounded web searches used the same terms together with rough- and multiplicative-surface-signature variants. No later explicit answer or this construction was located. This is a bounded novelty check, not an exhaustive bibliographic claim.

References

- [1] D. Lee, *The Surface Signature and Rough Surfaces*, arXiv:2406.16857v2 (2024).